

04_Technical_Specification

ALDRIDGE HARBOR PORT AUTHORITY

KESSLER POINT MARINE TERMINAL — PERIMETER SECURITY, ACCESS CONTROL AND SURVEILLANCE SYSTEMS

DOCUMENT 4 OF 13 — TECHNICAL SPECIFICATION

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SENSITIVE SECURITY INFORMATION — 49 CFR PART 1520

ABBREVIATIONS AND ACRONYMS

Term	Meaning
AES	Advanced Encryption Standard
AGC	Automatic Gain Control
AI	Artificial Intelligence
ANPR	Automatic Number Plate Recognition
ARO	After Receipt of Order
ASTM	ASTM International
BLC	Back Light Compensation
CBR	Constant Bit Rate
CHUID	Cardholder Unique Identifier
CMOS	Complementary Metal Oxide Semiconductor
DORI	Detect, Observe, Recognise, Identify (EN 62676-4)
EN	European Norm
FASC-N	Federal Agency Smart Credential Number
FHT	Full-Height Turnstile
FICAM	Federal Identity, Credential and Access Management
FIPS	Federal Information Processing Standard
FoV	Field of View
fps	Frames per second
GOP	Group of Pictures
GOS	Gate Operating System
H.264 / H.265	Video compression standards
HLC	Highlight Compensation
IEC	International Electrotechnical Commission
IK	Impact Protection rating (IEC 62262)
IP (rating)	Ingress Protection rating (IEC 60529)
IP (network)	Internet Protocol

IR	Infrared
MCBF	Mean Cycles Between Failures
MTBF	Mean Time Between Failures
NAR	Nuisance Alarm Rate
NEC	National Electrical Code (NFPA 70)
NFPA	National Fire Protection Association
ONVIF	Open Network Video Interface Forum
OSDP	Open Supervised Device Protocol
Pd	Probability of Detection
PIDS	Perimeter Intrusion Detection System
PoE	Power over Ethernet
PSIM	Physical Security Information Management
PTZ	Pan, Tilt, Zoom
RFID	Radio Frequency Identification
RTSP	Real Time Streaming Protocol
SC	Secure Channel (OSDP)
SNR	Signal to Noise Ratio
SOC	Security Operations Centre
SRTP	Secure Real-time Transport Protocol
TIA	Telecommunications Industry Association
TLS	Transport Layer Security
TOS	Terminal Operating System
TPM	Trusted Platform Module
TWIC	Transportation Worker Identification Credential
UL	UL Solutions (Underwriters Laboratories)
VBR	Variable Bit Rate
VMS	Video Management System
WDR	Wide Dynamic Range

1. GENERAL TECHNICAL REQUIREMENTS

TS-1.1 All equipment shall be new, unused, of current manufacture, and shall not be scheduled for end of production or end of support within five (5) years of Final Acceptance.

TS-1.2 All equipment installed externally shall be rated for the marine environment described at Annex A, including salt spray, wind-driven rain and chloride exposure, and shall be suitable for continuous exposure to atmospheric corrosivity category C5-M to ISO 9223.

TS-1.3 All equipment shall comply with the standards scheduled at Annex B.

TS-1.4 No equipment, system, component or service produced or provided by an entity named at ITB-12.2, or by any subsidiary or affiliate of such an entity, shall be supplied. This prohibition applies to original equipment manufacturer supply to another brand.

TS-1.5 The Contractor shall submit, for every proposed item, a declaration of the manufacturing origin of the device and of its principal sub-assemblies, including image sensor, video encoder and network interface.

TS-1.6 All systems shall be capable of operation at MARSEC Levels 1, 2 and 3, and shall support escalation of security measures at each level without hardware change.

TS-1.7 All systems will provide a means of recording operator actions for audit purposes.

TS-1.8 The systems shall present a unified operator interface. Operators shall not be required to use more than one application in normal operation.

TS-1.9 All externally mounted equipment shall be supplied with stainless steel grade 316 fixings. Dissimilar metal contact shall be isolated.

TS-1.10 All equipment shall be supplied with a manufacturer's warranty of not less than thirty-six (36) months.

2. SYSTEM A — PERIMETER INTRUSION DETECTION SYSTEM

2.1 General

TS-A.1 The Contractor shall supply and install a taut-wire perimeter intrusion detection fence to the full landside boundary of the Site, a length of approximately **8,200 linear feet**, together with an internal patrol road.

TS-A.2 The detection fence shall comprise not fewer than **twenty (20)** horizontal tensioned wires, anchored to detector posts at intervals not exceeding **ten (10) feet**.

TS-A.3 The system shall be divided into detection zones not exceeding **330 feet** in length.

TS-A.4 Each zone shall report independently to the Security Management System.

2.2 Detection performance

TS-A.5 The system shall achieve a Probability of Detection (Pd) of not less than **ninety-five percent (95%)** against the following defeat methods: climbing, cutting, spreading, and lifting of the wires.

TS-A.6 The system shall exhibit a Nuisance Alarm Rate (NAR) not exceeding **one (1) alarm per zone per twenty-four (24) hours**.

TS-A.7 Pd and NAR shall be demonstrated over a continuous thirty (30) day soak period during the Site Acceptance Test.

TS-A.8 The system shall have a Mean Time Between Failures of not less than **60,000 hours**.

TS-A.9 The system shall not be defeated by wind, rain, hail, wildlife or debris common to the Site.

TS-A.15 Wire tension shall be monitored continuously and a tension deviation exceeding the manufacturer's stated threshold shall raise a supervisory alarm.

TS-A.16 The system shall discriminate between an intrusion event and an environmental event, and shall report the discrimination decision to the operator.

TS-A.17 Detector post sensors shall operate over an ambient temperature range of -20°C to $+60^{\circ}\text{C}$ and at relative humidity to 100% non-condensing.

TS-A.18 Each zone shall be individually addressable for the purposes of inhibit, test and reset, and no inhibit shall persist beyond a configurable timeout without an operator confirmation.

2.3 Construction

TS-A.10 Detector posts shall be hot-dip galvanised to ASTM A123 and shall additionally be polyester powder coated.

TS-A.11 Foundations shall be designed for the wind loading stated at Annex A and for the soil conditions encountered.

TS-A.12 The patrol road shall be not less than **twelve (12) feet** wide, constructed of compacted aggregate, and shall run continuously inside the detection line.

TS-A.13 Where the detection line is interrupted by a gate, the detection shall be continued across the opening by a means proposed by the Contractor and approved by the Authority.

TS-A.14 Supplementary fibre-optic fence sensing shall be provided to the approximately **1,800 linear feet** of boundary identified on the drawings as visually obstructed.

TS-A.19 All field wiring between detector posts and zone controllers shall be screened, and the screen shall be earthed at one end only.

TS-A.20 All cable entries to detector posts and zone controllers shall be from below and shall be sealed against water and vermin ingress.

3. SYSTEM B — VIDEO SURVEILLANCE SYSTEM

3.1 General

TS-B.1 The Contractor shall supply and install a video surveillance system comprising approximately **174 cameras**, a video management system, recording, storage and analytics, as scheduled in the Bill of Quantities.

TS-B.2 All cameras shall be network cameras supporting **H.265 (Main Profile) and H.264 (Main and High Profile)** encoding, with a configurable Group of Pictures length.

TS-B.3 All cameras shall conform to **ONVIF Profile S, Profile G and Profile T**.

TS-B.4 All cameras shall support **FIPS 140-3 validated** cryptographic modules for the protection of video and control traffic.

TS-B.5 All cameras shall support **HTTPS with TLS 1.3, SRTP**, IEEE 802.1X certificate-based authentication, signed firmware, and a hardware root of trust implemented in a TPM or equivalent secure element.

TS-B.6 All cameras shall support simultaneous delivery of not fewer than **four (4)** independently configurable video streams, each with individually selectable resolution, frame rate, compression standard and bit rate control mode.

TS-B.7 All cameras shall support both **constant bit rate and variable bit rate** encoding, and a manufacturer's bandwidth-reduction technology shall be configurable independently per stream.

TS-B.8 All cameras shall provide edge storage by removable media of not less than **256 GB**, recording continuously on loss of network connection and automatically replenishing the recording gap at the head end on restoration.

3.2 Perimeter detection cameras — imaging

TS-B.10 Perimeter detection cameras shall employ a **progressive scan CMOS** image sensor of size not smaller than **1/2.8 inch**.

TS-B.11 Perimeter detection cameras shall provide a native resolution of not less than **1920 × 1080** pixels.

TS-B.12 Perimeter detection cameras shall deliver not fewer than **30 frames per second** at full native resolution, with all analytics enabled.

TS-B.13 Perimeter detection cameras shall be fitted with a **motorised varifocal lens of focal length range 2.8 mm to 12 mm**, with remote focus and remote zoom adjustable from the video management system.

TS-B.14 The lens shall have a maximum aperture of **F1.4 or faster**, and shall be of the P-iris or DC-iris type with automatic iris control.

TS-B.15 The lens shall be **IR-corrected** so that focus is maintained between day and night modes without refocusing.

TS-B.16 Perimeter detection cameras shall achieve a minimum illumination of not more than:

Mode	Illumination	Conditions
Colour	0.005 lux	F1.4, 1/30 s, 30 IRE
Monochrome	0.0005 lux	F1.4, 1/30 s, 30 IRE
Monochrome with IR illumination	0 lux	—

TS-B.17 Perimeter detection cameras shall provide **Wide Dynamic Range of not less than 140 dB**, measured in accordance with the manufacturer's published method.

TS-B.18 Perimeter detection cameras shall provide a signal-to-noise ratio of not less than **50 dB**.

TS-B.19 Perimeter detection cameras shall provide an electronic shutter adjustable over the range **1/6 s to 1/100,000 s**, with automatic and manual modes.

TS-B.20 Perimeter detection cameras shall provide back light compensation, highlight compensation, automatic gain control with a configurable maximum, 3D digital noise reduction, defog, and electronic image stabilisation.

TS-B.21 Perimeter detection cameras shall incorporate a mechanically switched infrared cut filter with automatic day and night changeover, and a manually selectable changeover threshold.

TS-B.22 Where infrared illumination is provided, it shall be of wavelength 850 nm, shall be synchronised to the lens zoom position, and shall provide a useful range of not less than **50 metres**.

3.3 Perimeter detection cameras — coverage and analytics

TS-B.25 Each perimeter detection camera shall achieve a pixel density of not less than **eighty (80) pixels per metre at the far edge of its detection zone**, calculated from the sensor width, the horizontal resolution and the lens focal length proposed. Detection zone extents are as shown on drawings SEC-201 and SEC-202.

TS-B.26 The Contractor shall submit, for every camera position, a field-of-view calculation demonstrating compliance with TS-B.25.

TS-B.27 Perimeter detection cameras shall provide artificial-intelligence object detection and classification, performed **at the edge**, distinguishing as a minimum between **person, vehicle, vessel and animal**.

TS-B.28 Classification accuracy shall be not less than **ninety-five percent (95%)**.

TS-B.29 Analytics shall support, as a minimum, the following rule types: line crossing with direction; intrusion into a defined region; loitering with a configurable dwell time; object left behind; object removed; crowd formation; and camera tamper, defocus and scene change.

TS-B.30 Analytics shall support not fewer than **eight (8)** independently configurable rules per camera, each with its own schedule and its own region of interest.

TS-B.31 The camera shall stream object metadata, including object class, bounding box, trajectory and dominant colour, as a separate stream to the video management system, for use in forensic search without re-analysing recorded video.

TS-B.32 Analytics shall continue to operate on loss of connection to the head end, and shall buffer events locally for transmission on restoration.

3.4 Perimeter detection cameras — environmental and power

TS-B.35 Perimeter detection cameras shall be rated **IP67** to IEC 60529 and **IK10** to IEC 62262.

TS-B.36 Perimeter detection cameras shall operate over an ambient temperature range of **−40 °C to +60 °C**.

TS-B.37 Perimeter detection cameras shall be supplied with an integral heater and blower, a sunshield, and a hydrophobic coating to the window.

TS-B.38 Perimeter detection cameras shall be powered by **IEEE 802.3af Power over Ethernet, not exceeding 15.4 W per device**, from the field cabinet serving that sector.

TS-B.39 Perimeter detection cameras shall have a Mean Time Between Failures of not less than **100,000 hours** at 25 °C.

3.5 Assessment cameras

TS-B.40 PTZ assessment cameras shall provide not less than **36× optical zoom** and a focal length range of not less than **6 mm to 220 mm**.

TS-B.41 PTZ assessment cameras shall provide continuous 360° pan, tilt from +20° to −90°, and not fewer than **300** presets.

TS-B.42 Pan speed shall be not less than 200°/s on preset recall and variable from 0.1°/s for manual control.

TS-B.43 PTZ assessment cameras shall move to a preset position on receipt of an alarm from any detection zone within **three (3) seconds**.

TS-B.44 PTZ assessment cameras shall provide a native resolution of not less than 1920 × 1080 at 30 fps, WDR of not less than 120 dB, and optical image stabilisation.

TS-B.45 PTZ assessment cameras shall support automatic tracking of a detected object, with handover to an adjacent camera where the object leaves the field of view.

TS-B.46 PTZ assessment cameras shall be rated **IP66** as a minimum, and shall incorporate a wiper to the window.

3.6 Waterside cameras

TS-B.50 Thermal imaging cameras shall be provided along the quay face, with a human detection range of not less than **1,200 metres** determined in accordance with the Johnson criteria.

TS-B.51 Thermal imaging cameras shall employ an uncooled vanadium oxide microbolometer of resolution not less than 640 × 512, with a pixel pitch not exceeding 17 µm and a thermal sensitivity (NETD) of not more than 40 mK.

TS-B.52 Thermal imaging cameras shall provide on-board analytics capable of classifying a detected object as person, vehicle or vessel.

TS-B.53 Thermal cameras will be integrated with the surface radar such that a radar track cues the nearest thermal camera.

3.7 Video management, recording and storage

TS-B.60 All video shall be recorded continuously at full frame rate and full resolution.

TS-B.61 Recorded video shall be retained for not less than **thirty (30) days**.

TS-B.62 Storage shall be configured with RAID-6 or equivalent protection, and shall tolerate the failure of any two drives without loss of recording.

TS-B.63 Storage shall be sized to provide not less than twenty percent (20%) spare capacity at Final Acceptance.

TS-B.64 The video management system shall support not fewer than **twelve (12)** concurrent operator clients.

TS-B.65 The video management system shall support not fewer than **500** cameras without additional server hardware, to permit future expansion.

TS-B.66 The video management system shall provide forensic search across all cameras by object class, colour, direction of travel and time, using the metadata stream at TS-B.31.

TS-B.67 Video export shall be in a format playable without proprietary software, with an accompanying integrity hash and a tamper-evident audit record of the export.

TS-B.68 The video management system shall record an audit entry for every video access, export and configuration change, attributable to an individual user.

TS-B.69 The video management system shall provide adequate video quality for the identification of individuals at all access control points.

TS-B.70 Recording shall not be interrupted by the failure of any single server, switch, power supply or network path.

TS-B.71 Lighting shall be sufficient to support the performance required by this specification.

4. SYSTEM C — SECURITY PRE-GATE

4.1 General

TS-C.1 The Contractor shall supply and install a security pre-gate comprising **four (4) inbound lanes**, an optical character recognition portal, credential

readers, barriers, and a Command and Control station.

TS-C.2 The pre-gate shall be located upstream of the Commercial Gate. The inter-gate queue lane is provided by others.

TS-C.3 There shall be no outbound pre-gate lanes.

4.2 Optical character recognition

TS-C.4 The OCR portal shall read vehicle licence plates with an accuracy of not less than **98%** at speeds up to **15 miles per hour**.

TS-C.5 The OCR portal shall read chassis identification numbers with an accuracy of not less than **95%**.

TS-C.6 Container number recognition is **not** required at the pre-gate.

TS-C.21 OCR accuracy shall be measured over a continuous seven (7) day period of normal operation, by comparison against a manually verified sample of not fewer than 500 transactions.

TS-C.22 The OCR portal shall be provided with dedicated illumination, synchronised to the capture, and shall operate through the full range of ambient conditions at Annex A including direct low sun on the approach.

TS-C.23 The OCR system shall return a confidence value with every read, and a read below a configurable confidence threshold shall be presented to the Command and Control operator for manual confirmation.

TS-C.24 The OCR system shall capture and retain an image of every plate and chassis read, linked to the transaction record.

4.3 Credential and vehicle identification

TS-C.7 Each lane shall be equipped with an RFID reader capable of reading the tractor tags in use at the Site.

TS-C.8 Each lane shall be equipped with a credential reader capable of reading a Transportation Worker Identification Credential.

TS-C.9 Credential readers shall be listed on the Transportation Security Administration's TWIC Reader Qualified Technology List and shall perform **biometric fingerprint verification** of the cardholder against the credential in accordance with the TWIC Reader Rule.

TS-C.10 Credential readers shall be listed on the FICAM Approved Products List maintained at idmanagement.gov.

TS-C.11 Each lane shall be equipped with a driver face-capture camera, a cab overview camera, and an intercom to the Command and Control station.

TS-C.25 The driver face-capture camera shall achieve a pixel density of not less than **250 pixels per metre** across the expected head position, corresponding to DORI Identify.

TS-C.26 Lane readers shall communicate with the lane controller using **OSDP v2.2 with Secure Channel enabled**. Wiegand shall be provided as a parallel output for compatibility with the existing system.

4.4 Transaction logic

TS-C.12 For each inbound vehicle, the system shall: read the licence plate; read the chassis identification; read the RFID tractor tag; read the driver credential; verify the credential against the TWIC Canceled Card List; query the Terminal Operating System to confirm that the driver and vehicle combination has a valid reason to enter; query the law enforcement watchlist interface; and present a decision to the Command and Control operator.

TS-C.13 The complete transaction at TS-C.12 shall be performed in not more than **twenty (20) seconds**.

TS-C.14 Where any element of the transaction fails, the vehicle shall be directed to the inspection lay-by.

TS-C.15 The system shall be capable of processing not fewer than **180 vehicles per lane per hour**.

TS-C.16 At MARSEC Level 2 the system shall support one hundred percent (100%) inspection without reduction in the throughput required by TS-C.15.

4.5 Command and Control station

TS-C.17 The Command and Control station shall provide **two (2)** operator positions with a view of all four lanes.

TS-C.18 Each position shall present, for the vehicle under decision, the plate image, the chassis image, the driver face capture, the credential data, the Terminal Operating System response and the watchlist response on a single screen.

TS-C.19 The operator shall be able to admit, divert or reject a vehicle by a single action.

TS-C.20 All decisions shall be logged with the operator identity, the decision and the time.

TS-C.27 The Command and Control station shall be provided with a duress facility at each operator position, raising a silent alarm at the Security Operations Centre.

5. SYSTEM D — ELECTRONIC ACCESS CONTROL

5.1 General

TS-D.1 The Contractor shall supply and install an electronic access control system serving approximately **1,800 credential holders** across all access points.

TS-D.2 All access control units shall be listed to **UL 294, 8th Edition**, at **Level IV** for destructive attack, line security, endurance and standby power.

TS-D.3 All access control system components shall be listed on the FICAM Approved Products List and shall comply with **FIPS 201-3**.

TS-D.4 The system shall support the TWIC as the primary credential, reading the CHUID and FASC-N contactlessly.

TS-D.5 The system shall check every credential presentation against the TWIC Canceled Card List, updated not less frequently than daily.

TS-D.6 The system shall support anti-passback at all controlled portals.

5.2 Readers

TS-D.20 Credential readers shall operate at **13.56 MHz** and shall support ISO/IEC 14443 Types A and B and ISO/IEC 7816.

TS-D.21 Credential readers shall communicate with controllers using **OSDP v2.2 with Secure Channel enabled**, using AES-128 encryption with per-device keys.

TS-D.22 Readers shall achieve a read range of not less than **50 mm** for a TWIC credential and a read time not exceeding **500 ms** from presentation to decision request.

TS-D.23 Readers installed externally shall be rated IP66 and IK09 as a minimum and shall operate over -20°C to $+60^{\circ}\text{C}$.

TS-D.24 Readers shall provide tamper detection, reporting both cover removal and removal from the mounting surface.

TS-D.25 Readers shall provide visual and audible indication of accept, reject and system fault states.

TS-D.26 Reader firmware shall be field-updateable over the OSDP connection, and shall accept only cryptographically signed firmware.

5.3 Controllers

TS-D.30 Access control panels shall support not fewer than **sixteen (16)** readers per panel.

TS-D.31 Panels shall hold not fewer than **200,000** credential records and not fewer than **500,000** offline event records.

TS-D.32 Panels shall continue to make access decisions and to record events with the head-end server unavailable, for a period of not less than **seventy-two (72) hours**, and shall upload buffered events automatically on restoration.

TS-D.33 Panels shall be supplied with a battery providing not less than **four (4) hours** of standby operation, with battery condition monitored and reported.

TS-D.34 Panel enclosures shall be provided with tamper detection on the door and on removal from the wall.

TS-D.35 All panel field wiring shall be supervised, and an open, short or earth fault on any input shall be reported as distinct from an alarm condition.

TS-D.36 Panels shall support elevated threat-level operation, by which a single command changes the access rules applied at every portal to those defined for MARSEC Levels 2 and 3.

5.4 Pedestrian entry

TS-D.7 The Contractor shall supply and install **six (6)** full-height turnstile lanes, arranged as four lanes at the main personnel entry and two at the labour hall entry.

TS-D.8 Full-height turnstiles shall be constructed of stainless steel to grade 316 and shall be suitable for continuous marine exposure.

TS-D.9 Each turnstile lane shall be fitted with credential readers on both the entry and exit sides.

TS-D.10 Each turnstile lane shall achieve a throughput of not less than **ten (10) persons per minute**.

TS-D.11 Turnstiles shall fail to the free-rotation condition on loss of power and on receipt of a fire alarm signal.

TS-D.12 The Contractor shall supply and install **three (3)** accessible swing gates, one at each turnstile bank and one at the gangway, each with credential readers on both sides.

TS-D.13 Each turnstile lane shall be provided with an overhead camera and a face-capture camera in each direction.

TS-D.40 Turnstile rotors shall be of four-arm configuration with a 90° segment, and shall provide a clear passage width of not less than **660 mm** and a clear height of not less than **2,200 mm**.

TS-D.41 Turnstile mechanisms shall be rated for not fewer than **two million (2,000,000)** cycles, with a Mean Cycles Between Failures of not less than 1,000,000.

TS-D.42 Turnstiles shall incorporate a hydraulic or electromechanical damper preventing free-spin and controlling rotor deceleration.

TS-D.43 Turnstiles shall provide anti-tailgate control by mechanical segment, permitting the passage of one person per authorised presentation.

TS-D.44 Turnstiles shall provide a directional indication to the user, and shall lock against movement in the unauthorised direction.

TS-D.45 Turnstile lanes shall be supplied with integral lighting and shall be finished to withstand chloride exposure without visible corrosion for not less than ten (10) years.

5.5 Portals and door hardware

TS-D.50 Electric locking devices shall be fail-secure at all external portals and fail-safe on designated egress routes, as scheduled on drawing SEC-404.

TS-D.51 Electromagnetic locks, where used, shall provide a holding force of not less than **1,200 lbf** and shall be monitored for bond status.

TS-D.52 All controlled doors shall be monitored for position, forced entry and door-held-open, each reported as a distinct event.

TS-D.53 Request-to-exit devices shall be provided at all controlled doors and shall not be the sole means of egress.

5.6 Software

TS-D.60 The access control application shall provide role-based administration, with the ability to delegate cardholder administration without granting configuration rights.

TS-D.61 The system shall support scheduled and event-driven changes to access rules, and shall support temporary credentials with an automatic expiry.

TS-D.62 The system shall support visitor management, including pre-registration, escort assignment and automatic expiry.

TS-D.63 The system shall record, for every credential holder, the TWIC expiry date, and shall automatically deny access on expiry without administrator action.

TS-D.64 The system shall provide a documented application programming interface for integration to the PSIM platform and to the Authority's identity systems.

TS-D.65 The system database shall be replicated to a standby instance with automatic failover, and the failover shall be demonstrated at the Site Acceptance Test.

6. SYSTEM E — VEHICLE BARRIERS AND GATES

TS-E.1 The Contractor shall supply and install **five (5)** crash-rated sliding gates and **twelve (12)** vertical arm barriers as scheduled in the Bill of Quantities.

TS-E.2 Crash-rated sliding gates shall be certified to **ASTM F2656 rating M50 / P1**.

TS-E.3 Crash-rated sliding gates at two-lane crossings shall provide a clear opening of not less than **twenty-four (24) feet**. Single-lane gates shall provide a clear opening of not less than sixteen (16) feet.

TS-E.4 Crash-rated sliding gates shall achieve a full open or full close cycle in not more than **ten (10) seconds**.

TS-E.5 All gate operators shall be listed to **UL 325** and all automated vehicular gates shall comply with **ASTM F2200**.

TS-E.6 Gate operators shall be provided with entrapment protection in accordance with UL 325 for the applicable usage class.

TS-E.7 Vertical arm barriers shall achieve a full open or full close cycle in not more than **1.5 seconds**.

TS-E.8 Vertical arm barriers shall be rated for not fewer than **three million (3,000,000)** operating cycles.

TS-E.9 All gates and barriers shall be provided with local manual release for emergency access.

TS-E.10 All gates and barriers shall be integrated to the Security Management System such that position and fault status are reported.

TS-E.11 Barrier arms shall be of a frangible or articulating type such that impact does not damage the operator mechanism.

TS-E.12 Each barrier shall be provided with a safety loop and a photoelectric beam preventing closure on a vehicle.

TS-E.13 Gate and barrier operators shall be supplied with a battery backup providing not fewer than fifty (50) cycles or two (2) hours of standby, whichever is the greater.

7. SYSTEM F — SECURITY OPERATIONS CENTRE

TS-F.1 The Contractor shall fit out a Security Operations Centre of approximately **2,200 square feet**, comprising an operations floor, an equipment room, a Facility Security Officer's office and support accommodation.

TS-F.2 The operations floor shall accommodate **four (4)** operator consoles.

TS-F.3 Each console shall be provided with not fewer than four (4) displays.

TS-F.4 A video wall of **fifteen (15)** displays in a three by five array shall be provided.

TS-F.5 The video wall shall be driven by a controller supporting operator-selectable layouts and alarm-driven automatic presentation.

TS-F.6 The equipment room shall accommodate **six (6)** 42U equipment racks.

TS-F.7 The equipment room shall be provided with dedicated cooling sized for the connected load plus fifty percent (50%), with N+1 redundancy.

TS-F.8 A Physical Security Information Management platform shall be provided, presenting all systems in a single operator interface with alarm handling, standard operating procedures and audit logging.

TS-F.9 The operations floor shall have a background noise level not exceeding **35 dBA**.

TS-F.10 The operations floor shall be provided with lighting appropriate to continuous screen-based work.

8. SYSTEM G — NETWORK AND CABLING

TS-G.1 The Contractor shall supply and install a dedicated security network, physically separate from the terminal operational technology network and from the Authority's business network.

TS-G.2 The security network backbone shall be single-mode fibre in a **fully redundant ring topology serving every field cabinet**.

TS-G.3 Core switching shall be provided in a redundant pair with sub-second failover.

TS-G.4 All field-connected devices shall authenticate to the network using **IEEE 802.1X**.

TS-G.5 The Contractor shall supply and install **twelve (12)** outdoor field cabinets, rated NEMA 4X, of stainless steel construction, thermostatically controlled, each containing a PoE switch, fibre termination and a local uninterruptible power supply.

TS-G.6 Structured cabling shall comply with TIA-568, TIA-569 and TIA-606.

TS-G.7 All outdoor cabling shall be provided with surge protection at both ends.

TS-G.8 All cables shall be labelled at both ends in accordance with TIA-606 and shall be recorded in the as-built cable schedule.

TS-G.9 Field switches shall provide PoE budget sufficient for all connected devices with one spare port of full budget per cabinet.

TS-G.10 The network shall be designed with not less than fifty percent (50%) spare bandwidth on every backbone link at the design camera count and bit rate.

9. SYSTEM H — POWER

TS-H.1 The Contractor shall supply and install **two (2)** uninterruptible power supplies of **40 kVA** each, configured N+1.

TS-H.2 The uninterruptible power supplies shall sustain the full connected security load for not less than **thirty (30) minutes**.

TS-H.3 The Contractor shall supply and install a standby generator of not less than **250 kW**, compliant with **NFPA 110**, with a fuel tank sized for **seventy-two (72) hours** at full load.

TS-H.4 The generator shall start and accept load within **ten (10) seconds** of loss of utility supply.

TS-H.5 Each outdoor field cabinet shall be provided with a local uninterruptible power supply sustaining the cabinet load for not less than **thirty (30) minutes**.

TS-H.6 All security power distribution shall be separately metered and monitored.

10. SYSTEM I — CIVIL AND BUILDER'S WORK

TS-I.1 The Contractor shall provide all civil works necessary for the Works, including foundations for poles, masts, cabinets, gates, barriers and turnstiles; ducts and chambers; the patrol road; and reinstatement of all disturbed surfaces.

TS-I.2 All foundations shall be designed by a Professional Engineer registered in the State.

TS-I.3 All excavation shall be preceded by utility location and by hand digging within three (3) feet of any known service.

TS-I.4 All reinstatement shall match the existing surface in construction and finish.

TS-I.5 The Contractor shall make good any damage to terminal pavements, rails, services or equipment arising from its works.

END OF DOCUMENT 4